

Single-Platform Nanopore Sequencing Enables Diploid Telomere-to-Telomere Genome Assembly and Haplotype-Resolved 3D Chromatin Maps

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Abstract

Telomere-to-telomere (T2T) genome assembly has transformed human genomics by resolving centromeres, segmental duplications, and other previously inaccessible regions. However, most diploid T2T assemblies rely on the combination of multi-platform sequencing strategies including short read genome sequencing, PacBio HiFi, Oxford Nanopore ultra-long (UL) reads, and chromatin conformation capture data (Hi-C), limiting both scalability and accessibility. Here, we present a streamlined Nanopore-only workflow for diploid human T2T assembly using three UL and one Pore-C PromethION flow cell per individual. Across 23 genetically diverse individuals, we generated 360 gapless chromosomes and 446 near-complete T2T scaffolds. Additional assembly of HG002 revealed a median consensus accuracy of QV68 in non-repetitive sequence and QV41 genome-wide including satellite arrays without Duplex sequencing or hybrid polishing. Assembly continuity, gene completeness, and structural variant detection were comparable to multi-platform Human Pangenome Reference Consortium assemblies. Pore-C data enabled chromosome-scale haplotype phasing without parental information and generation of haplotype-resolved chromatin contact maps, while methylation patterns at imprinted loci allowed assignment of parental origin. Integrated methylation and 3D genome analyses revealed allele-specific chromatin organization at imprinted loci and clear signatures of X-chromosome inactivation. For high-resolution chromatin contact mapping, additional Pore-C flow cells were generated. Our openly accessible dataset expands public T2T resources and demonstrates that reference-grade diploid assemblies, phased methylomes, and 3D genome maps can be derived from a single sequencing platform. This approach reduces technical barriers and supports scalable population and functional genomics in the T2T era.

56 Introduction

57 The recent completion of a telomere-to-telomere (T2T) human reference genome¹ and the
58 subsequent assembly of the first fully phased diploid human genome² marked a milestone in
59 genomics, resolving previously inaccessible regions including centromeres, segmental duplications,
60 and large repetitive arrays. This achievement lays the foundation for the Human Pangenome
61 Reference Consortium (HPRC), which aims to construct a comprehensive human pangenome based
62 exclusively on high-quality T2T assemblies. The initial pangenome draft ³ has already demonstrated
63 the value of multiple complete assemblies for resolving structural variation, characterizing
64 centromere architecture⁴, and improving variant discovery in complex genomic regions⁵. Population-
65 specific T2T assemblies further highlight the importance of complete references for understanding
66 genomic diversity across a wide range of ancestries ^{6,7}.

67 Current strategies for generating diploid T2T assemblies typically rely on the integration of multiple
68 sequencing platforms using assembly methods such as *Verkko*⁸ or *hifiasm*⁹. High accuracy long reads,
69 most commonly PacBio HiFi, are used to construct the primary assembly graph, while Oxford
70 Nanopore Technologies (ONT) ultra-long (UL) reads provide spanning information to resolve repeats
71 and close remaining gaps. Additional datasets such as Hi-C¹⁰, Pore-C, Strand-seq¹¹, or parental trio
72 sequencing are then employed for haplotype phasing and scaffolding. While highly effective, these
73 multi-platform approaches are **cost-intensive** and come with increased-laboratory complexity and
74 logistical requirements, thus limiting broad accessibility outside large genome centers. Recently,
75 Nanopore-only strategies have been explored as a potential alternative. Near-complete assemblies
76 have been generated without **UL** reads¹², and gapless assemblies have been demonstrated using
77 combinations of **UL** reads, Pore-C, and Nanopore Duplex sequencing¹³. Duplex reads achieve
78 improved per-read accuracy by combining signals from both DNA strands; however, their yield
79 remains limited and the workflow adds additional complexity and cost. Thus, **whether the**
80 **combination of UL reads and Pore-C alone could yield high-quality diploid T2T assemblies at**
81 **population scale, without HiFi data and Duplex sequencing, had not been systematically**
82 **demonstrated.**

83 Beyond linear genome reconstruction, comprehensive diploid assemblies provide an opportunity to
84 integrate higher-order genome organization and epigenetic information. The three-dimensional (3D)
85 chromatin architecture of the genome is organized into topologically associating domains (TADs)¹⁴,

which constrain regulatory interactions and are frequently demarcated by CTCF binding sites and cohesin-mediated loop extrusion^{15, 16}. While chromatin structure is dynamically re-established during the cell cycle, allele-specific TAD configurations can arise through epigenetic mechanisms such as imprinting-mediated differential methylation at CTCF binding sites¹⁷. Chromatin conformation capture (3C) technologies, particularly Hi-C¹⁰, have enabled genome-wide interrogation of chromosomal contacts and are routinely used for scaffolding and phasing in *de novo* assemblies. Pore-C extends this principle by coupling proximity ligation with long-read Nanopore sequencing, thereby capturing multi-way contacts within single concatemers¹⁸. Compared to short-read Hi-C, Pore-C enables higher-resolution mapping of mid-range chromatin interactions and provides improved long-range scaffolding power. When combined with phased heterozygous variants, Pore-C data allow assignment of contacts to individual haplotypes¹⁹, enabling haplotype-resolved maps of 3D genome organization²⁰. Importantly, Nanopore sequencing simultaneously preserves native DNA methylation information²¹, allowing joint reconstruction of phased genome sequence, chromatin topology, and methylome from a single technological platform.

Here, we demonstrate that high-quality, near-complete diploid human T2T assemblies can be generated using only Nanopore sequencing on four PromethION R10.4.1 flow cells, without PacBio HiFi data and without Duplex reads. Using three flow cells of UL reads and one flow cell of Pore-C data, we assembled a median of 16.5 out of 46 chromosomes gaplessly (range of 0 to 31) and achieved assembly quality metrics comparable to hybrid Nanopore–HiFi approaches. In parallel, we generated haplotype-phased maps of chromatin architecture and DNA methylation. Our streamlined workflow ~~substantially~~ reduces cost, laboratory complexity, and hands-on time while delivering integrated structural and functional genome information, providing a scalable framework for population-scale T2T genomics.

Results

A streamlined Nanopore-only workflow enables diploid T2T assembly from four flow cells

We generated diploid telomere-to-telomere (T2T) assemblies for 23 individuals, including two parent-child trios of German ancestry, using a Nanopore-only workflow (Fig. 1). Our standard operating procedure for T2T assembly uses three PromethION R10.4.1 flow cells of UL reads and one ~~to two~~ flow cells of Pore-C data (>50Gb yield), resulting in a total of four flow cells per genome. A

single Pore-C flow cell was sufficient for haplotype phasing and scaffolding in all samples. Additional Pore-C flow cells were sequenced for all samples to enable high-resolution chromatin contact mapping; however, these data were not used for assembly or phasing. The four-flow-cell target was consistently met for samples T2T05 - T2T19 after ultra-high molecular weight (UHMW) DNA extraction and UL library preparation had been optimized on samples T2T00 - T2T04 (Supplementary Table 2; Supplementary Fig. 1). No PacBio HiFi, Nanopore Duplex reads, Strand-seq, or parental data were required for assembly or haplotype phasing.

To ensure optimal long-read performance, we refined UL library preparation to maximize reads exceeding 100 kb while maintaining high total yield (Supplementary Fig. 2). UL reads were bioinformatically error-corrected using *HERRO* to generate high-quality (HQ) reads, which were used as substitutes for HiFi reads in *Verkko* assembly. Reads were filtered to retain 50× HQ coverage and 70× UL coverage, prioritizing read length and retaining only reads longer than 80kb to maximize repeat resolution (Supplementary Fig. 3). Pore-C data were preprocessed using ONT's *wf-pore-C* pipeline.

All data processing steps were automated combining preprocessing, assembly, phasing, quality control, variant calling, methylome and 3D genome analysis. Assemblies were generated using *Verkko* v2.2.1 with Pore-C-based or Trio-based phasing. Furthermore, we generated haplotype-phased assemblies using the *hifiasm* method for benchmarking purposes. For each assembler and phasing combination, we generated a wide range of quality measures to demonstrate the performance of Nanopore-only compared to hybrid-platform T2T assemblies. Computational costs of the workflow in terms of runtime as well as memory, CPU and GPU usage are shown for each step in Supplementary Fig. 4. On average, the runtime is 6 - 7 days, including super-accuracy base calling (40h), preprocessing (24h), *HERRO* correction (30h), and *Verkko* assembly and scaffolding (64h), which is comparable to a HiFi + UL + HiC assembly procedure.

High rates of gapless chromosome assembly across 23 diploid genomes

We classified *Verkko*-assembled chromosomes as T2T-scaffolds if they spanned ≥95% of the reference chromosome and contained telomeric repeats at both ends. Gap-free T2T-scaffolds were further classified as T2T contigs. The best-performing genome (T2T17) achieved 31 gapless T2T-contigs and 8 additional near-complete T2T-scaffolds (Fig. 2B). Across all 23 individuals, we

reconstructed 360 gapless chromosomes (34% of all haplotypes) and an additional 446 near-complete T2T-scaffolds (42%) (Supplementary Fig. 5). Many remaining gaps were small (<50 kb) and occurred as single isolated regions within otherwise continuous scaffolds.

Autosomal chromosomes 1–12 and 16–20 were consistently resolved to T2T contig or scaffold level across individuals. As expected, acrocentric chromosomes (13, 14, 15, 21, and 22) were more challenging; however, complete assemblies were achieved for specific acrocentrics in multiple individuals (e.g., chromosome 15 in T2T18; chromosome 22 in T2T04 and T2T16). In individual T2T18, all chromosomes except chromosome 13 were assembled to T2T scaffold or contig quality, including chromosome Y. **ChrY was resolved to T2T contig or scaffold level in 8 of 12 male samples and the HG002 cell line, with lengths ranging from 42 to 61 Mb (Supplementary Fig. 6), spanning the reported inter-individual variation in Y chromosome length²².**

Performance improved over time as UHMW DNA and UL library preparation was optimized (Fig. 2C), with later samples achieving higher T2T-contig counts independent of total sequencing coverage (Supplementary Fig. 5). Instead, the proportion of reads exceeding 100 kb strongly correlated with assembly continuity (Supplementary Fig. 3, 5).

Nanopore-only assemblies achieve hybrid-grade reference-free quality metrics

We next evaluated reference-free assembly quality metrics using *paftools* and compared our Nanopore-only assemblies to multi-platform (PacBio + ONT) assemblies from HPRC. Across genome size, gene completeness, missing multi-copy genes (MMC), retained single-copy genes (MSC), and T2T-contig counts, our assemblies displayed distributions comparable to HPRC assemblies (Fig. 2A; Supplementary Fig. 7). Underperforming samples were primarily from early protocol iterations, supporting the importance of optimized UHMW DNA isolation. **A consistent sex-related difference is apparent in several QC metrics due to reduced assembled length, reference-covered fraction, and completeness of chrY, reflecting both the intrinsic difficulty of assembling chrY and its structural divergence from the reference.**

To detect potential structural mis-assemblies, we re-aligned HQ reads and analyzed coverage and allele-frequency patterns using *NucFlag*²³. Assemblies with high T2T-contig counts showed correspondingly low numbers of flagged regions (Fig. 2C). Independent validation using *Flagger* confirmed these findings, with flagged regions enriched in known difficult genomic loci

(Supplementary Fig. 8, 9). To characterize the residual fragmentation, we classified all small contigs (10 kb–10 Mb) across all samples by satellite content and measured the distance from contig ends and assembly gaps to the nearest multicopy gene locus (Supplementary Fig. 10, 11). Satellite sequences, predominantly from centromeric and pericentromeric regions, account for 41.8% of small contigs, while a median of 36 non-satellite contigs (range: 4 - 378) were identified per sample. The remaining assembly discontinuities and gaps are highly enriched for direct overlap or very close proximity (within 10kb) to multicopy gene loci (41.8% and 68.5% respectively, random expectation: mean = 21.4%, 95% range = [20.2%, 22.3%], enrichment: 2 - 3.5-fold, $p < 0.005$).

Base-level consensus accuracy

Consensus accuracy across the cohort of 23 samples was first evaluated using *Mercury* based on Illumina short-read k-mers generated from >30x coverage per individual. For $k=31$, median QV across all samples was 51, corresponding to approximately 6 errors per megabase. Across k-mer sizes (21, 31, 37), median QV ranged from 48 to 60 (Supplementary Fig. 12A). However, reference-free k-mer analysis tends to overestimate accuracy in repetitive sequence, where errors are recurrent across reads (e.g. in homopolymers). We therefore assembled the genome of the HG002 cell line, for which high quality (Q100) reference genome is available, using our standard workflow (three UL and one Pore-C flow cell) and assembly pipeline. Comparison of the new HG002 assembly against the GIAB Q100 HG002v1.1 reference using *GQC* showed similar continuity on scaffold level (Supplementary Fig. 14A).

Base-level errors in HG002 were concentrated in low-complexity sequence (Supplementary Fig. 13, 14B). Of all discrepancies, 89.5% fell in homopolymer tracts and another 10.3% in other repeat classes. Non-repetitive sequence contained 781 errors in 4.83 Gb (QV 67.9) and coding sequence 56 errors in 70.3 Mb (QV 61.0). Pooled across all contexts, including satellite arrays, genome-wide QV was 40.7.

Indels accounted for 97.4% of errors and substitutions for 2.6% (about one transition and one transversion per megabase; QV 56.5). Indel errors were dominated by single-base events in homopolymers longer than 10bp (Supplementary Fig. 14C, 14D). The residual error profile reflects the known error patterns of Nanopore sequencing with R10.4 chemistry and is therefore not an artifact introduced by HERRO correction or other computational procedures.

206 Mendelian inheritance analysis of the two trio assemblies yielded *de novo* error rates of 2.2–2.7% at
207 evaluated heterozygous sites, corresponding to a genome-wide Mendelian QV of 48.8–49.5
208 (Supplementary Table 3), slightly above estimates of Merqury k-mer analysis of QV 44 - 45.

209 Trio-based phasing did not significantly improve QV. Integration of ONT's Assembly Polishing Kit APK,
210 using an additional flow cell and a separate library prep, did not improve reference free consensus
211 accuracy measured by Merqury (Supplementary Fig. 12B). APK was not used for the HG002 assembly.

212

213 **Pore-C–based phasing matches trio phasing performance**

214 Haplotype phasing was performed using Pore-C data for all samples. For two trios (T2T00, T2T04),
215 we compared Pore-C–based phasing to trio-based phasing. Both approaches generated
216 chromosome-scale haplotype blocks with consistent parental assignment across entire
217 chromosomes (Supplementary Fig. 15). Using Pore-C, the assignment of paternal or maternal origin
218 is arbitrary, but can be resolved using methylation-based parent-of-origin assignment
219 (Supplementary Fig. 18) as described below. Switch error rates and scaffold continuity were
220 comparable between Pore-C and trio phasing using Verkko. In contrast, assemblies generated with
221 hifiasm showed increased switch errors and reduced scaffold continuity relative to Verkko, despite
222 producing slightly higher T2T-contig counts (Supplementary Fig. 15, 16, 17). These results
223 demonstrate that Pore-C alone is sufficient for accurate haplotype phasing in Nanopore-only T2T
224 assemblies.

225

226 **Methylation-based parent-of-origin assignment resolves haplotype identity without parental data**

227 We used Nanopore UL-derived methylation data to infer the parent-of-origin for haplotypes of all
228 non-trio samples. To this end, we implemented an adapted version of PatMat²⁴, which infers parent-
229 of-origin using known imprinted differentially methylated regions (DMRs). While in the original
230 implementation chromosome-scale variant phasing is supplied by strand-seq data, we use Pore-C-
231 phased variants, so that no additional sequencing is required. Parent-of-origin was assigned for all
232 but six autosomes across the complete cohort (Supplementary Fig. 18). Three out of six unassigned
233 haplotypes occurred in chromosome 17, reflecting the limited number of informative imprinted
234 DMRs on this chromosome. In both trios, methylation-based assignment reproduced the parental

origin determined by trio-based phasing (Supplementary Fig. 15). Final assemblies are released with parent-of-origin-annotated haplotypes following the PanSN naming specification.

Diploid assemblies enable accurate phased variant discovery and ancestry inference

We aimed to confirm that the assembled genomes reflect the genetic variability expected from our cohort of diverse ancestries (Fig. 3A). We aligned each diploid assembly to T2T-CHM13 v2 to detect phased small and structural variants. For chromosomes assembled as T2T contigs or scaffolds, phase blocks spanned entire chromosomes.

On average, we identified 4.4 million single nucleotide variants (SNVs) per individual. Transition/transversion ratios and heterozygosity rates were comparable to non-African HPRC genomes (Fig. 3C; Supplementary Fig. 19A). A modest increase in small indels (1–5 bp) was observed, reflecting residual homopolymer-associated errors. Structural variant counts and chromosomal distributions closely matched HPRC samples (Fig. 3D), with ~15% homozygous structural variants (SVs) and balanced heterozygous SV distribution across haplotypes. **Chromosome 9 showed a markedly elevated SV count largely originating from the pericentromeric heterochromatin (9q12). 79.9% of chr9 SV calls fell within the CenSat-annotated centromere/satellite interval (chr9:40.0–81.7 Mb), where SV density is approximately 15-fold higher than in the chromosome arm (Supplementary Fig. 20). These calls largely reflect divergence in the length of heterochromatic arrays of our samples compared to the CHM13 reference rather than genuine excess variation.**

Principal component analysis using a T2T-lifted 1000 Genomes reference panel confirmed accurate ancestry clustering consistent with self-reported origins (Supplementary Fig. 21). Local ancestry inference of the assembled genomes, performed with RFMix using the coordinate lifted 1000 GP reference panel, confirmed the reported ancestries (Fig. 3B). Mixed ancestry was observed in only two samples with shared Admixed American (AMR) and European (EUR) ancestries, whereas all remaining samples match their expected 1000 Genomes **continental group** assignments.

Haplotype-resolved Pore-C reveals allele-specific chromatin architecture

Using phased SNVs, we haplotagged Pore-C reads and generated haplotype-resolved contact maps. Approximately 50% of Pore-C reads could be assigned to a haplotype, yielding up to 1.6 billion phased

264 contacts per sample. The assignment rate is limited by the density of phased SNVs and therefore
265 favors longer reads. Haplotype-assigned reads contributed up to 83% of contact pairs (Fig. 4C,
266 Supplementary Table 4). Beyond the single flow cell used for assembly and phasing, a second Pore-C
267 flow cell was sequenced for all samples to increase contact density. For female samples T2T12 and
268 T2T13 and male samples T2T00 and T2T04, additional Pore-C flow cells were generated to provide
269 higher resolution. The sequencing statistics and quality metrics of Pore-C flow cells are presented in
270 Supplementary Fig. 22 and 23, respectively.

271 Global chromatin architecture was highly concordant between haplotypes; however, allele-specific
272 differences were detectable in specific loci. At the imprinted *IGF2-H19* locus, haplotype-specific TAD
273 boundaries corresponded to differential methylation and CTCF binding (Fig. 4A). In samples of female
274 participants, X-chromosome inactivation was evident as global differences in contact density and
275 methylation between active and inactive X chromosomes (Fig. 4B).

276 These results demonstrate simultaneous reconstruction of phased genome sequence, methylome,
277 and 3D chromatin organization from a single sequencing platform.

278

279 **Nanopore-only assemblies reconstruct complete centromeric structures**

280 Centromeres are notoriously hard to assemble correctly due to large clusters of alpha satellite (α SAT)
281 tandem repeats²⁵. We characterized centromeric regions using CenMap. The number of fully
282 resolved centromeres per individual ranged from 2 to 42 (Fig. 5B), with highest resolution observed
283 in assemblies with the greatest UL read length distribution.

284 Consistent with prior studies, certain centromeres (e.g., chromosome 2) were reliably resolved,
285 whereas others (e.g., chromosome 18) remained challenging (Fig. 5A). Extensive variation in
286 centromere length and higher-order repeat (HOR) organization was observed between individuals
287 (Fig. 5C), highlighting the rapid evolutionary dynamics in centromeric regions²⁶.

288 The T2T18 haplotype 2 chromosome 5 centromere displayed a modular HOR architecture with
289 defined repeat families and orientations. CpG methylation and Pore-C mapping revealed epigenetic
290 heterogeneity and spatial clustering of repeats, respectively, while nucleotide identity analysis
291 showed mostly homogeneous sequences interspersed with variable regions. Together, these data
292 highlight that centromere structure and function arise from the coordinated interplay of sequence,
293 epigenetic modifications, and 3D genome organization (Fig. 5D).

294

295 Discussion

296 Telomere-to-telomere (T2T) assembly of diploid human genomes has thus far relied on the
297 integration of multiple sequencing technologies, typically combining PacBio HiFi reads for high
298 consensus accuracy, Oxford Nanopore UL reads for repeat resolution, and chromatin conformation
299 capture data from short-read platforms (Hi-C) for scaffolding and phasing. Here, we demonstrate
300 that high-quality diploid T2T assemblies can be generated using ONT as a sole sequencing platform
301 with two library types, UL and Pore-C, whereas comparable diploid T2T workflows combine PacBio
302 HiFi with ONT UL reads and Hi-C, Strand-seq, or parental trio sequencing for phasing (at least three
303 library types on up to three platforms). Our streamlined approach reduces infrastructure
304 requirements, consumable cost and hands-on laboratory work, but comes at the cost of a slightly
305 increased computational runtime due to HERRO error correction (around 30h). Overall, compute
306 requirements remain modest and are comparable to the multi-platform approach.

307 Across 23 individuals, we reconstructed 360 fully gapless chromosomes and an additional 446 near-
308 complete T2T scaffolds with four flow cells per genome for assembly and phasing using our optimized
309 protocol as demonstrated for samples T05 – T19 (Supplementary Table 2). As shown before for a
310 single genome¹³, we demonstrate that Pore-C data alone is sufficient for chromosome-scale
311 haplotype phasing. In direct comparison with trio-based phasing, Pore-C phasing produced similar
312 switch error rates and scaffold continuity, without requiring parental sequencing or additional
313 technologies such as Hi-C or Strand-seq on short-read platforms. This substantially simplifies diploid
314 genome assembly workflows and enables scalable phasing in cohorts where parental material is
315 unavailable. Furthermore, native methylation information from Nanopore reads enabled parent-of-
316 origin assignment of haplotypes using imprinted DMRs.

317

318 A key advance of this work is the reconstruction of diploid genomes reaching QV 68 in non-repetitive
319 regions and QV 41 genome-wide including satellite regions without Duplex sequencing or external
320 polishing, evaluated against the GiaB HG002 reference. Our results indicate that HERRO error-
321 corrected R10.4 UL reads can substitute for PacBio HiFi reads in graph-based assembly frameworks
322 such as Verkko, although an increased error rate in homopolymer stretches remain an unsolved issue
323 and largely account for the reduced genome-wide QV. While hybrid assemblies remain the current

324 gold standard, our results show that Nanopore-only assemblies can achieve similar gene
325 completeness, structural integrity, and phased variant detection performance.

326 Despite these advances, limitations remain. Base-level accuracy of Nanopore-only assemblies,
327 although approaching QV50, remains modestly below that achieved with HiFi-based assemblies,
328 particularly in homopolymeric regions where short indel errors persist. While polishing strategies
329 (e.g. APK) may further reduce these errors, our results suggest diminishing returns relative to
330 increased sequencing costs. Continued improvements in pore chemistry, base calling models,
331 computational error correction models and assembly algorithms will likely further narrow this gap.
332 Assembly continuity was highest for metacentric and submetacentric chromosomes, whereas
333 acrocentric chromosomes and certain centromeres remained challenging. Although complete
334 resolution of individual acrocentric chromosomes was achieved in selected samples, repetitive
335 ribosomal DNA arrays and large satellite blocks continue to represent the most difficult regions for
336 automated assembly.

337 Multi-copy genes are a well-known cause of contig breaks. Indeed, the residual contig ends and gaps
338 in our assemblies (including small non-satellite contigs) are strongly enriched for multicopy gene loci
339 (Supplementary Fig. 10). In addition, around 42% of small contigs originate from satellite regions, a
340 principal challenge for completing T2T assemblies. These limitations are not specific to our workflow
341 but common to current assembly and evaluation methods.

342 Advances in UHMW DNA extraction, further increases in read length distribution, and improved
343 repeat-aware assembly algorithms may ultimately resolve these remaining gaps. Indeed, we
344 observed a steady increase in assembly quality over the course of this study, which is mainly
345 attributed to the increase in yield of reads longer than 100 kb.

346 Importantly, we showed that the use of a single sequencing platform enables simultaneous
347 reconstruction of genome sequence, haplotype phase, methylation patterns, and 3D chromatin
348 organization. By integrating phased SNVs with Pore-C contact data, we generated haplotype-resolved
349 maps of TADs, revealing allele-specific chromatin structures at imprinted loci and clear signatures of
350 X-chromosome inactivation. Nanopore reads are more efficient for phased applications than Hi-C
351 reads for two reasons: their longer fragments span more heterozygous variants, improving phasing;
352 and individual reads capture multi-way contacts, increasing informative linkage. Together, these
353 features improve haplotagging and reduce the sequencing required to resolve allele-specific

chromatin structure at TAD-scale resolution compared with Hi-C-based approaches. This integrated framework extends beyond structural genome assembly and provides a foundation for functional diploid genomics.

The availability of phased T2T assemblies opens new opportunities for studying structural variation, centromere evolution, regulatory architecture, and population diversity. While short-read whole-genome sequencing captures approximately 93% of the human genome²⁷, unresolved regions are enriched for structural variants, satellite repeats, and regulatory domains that may contribute to rare diseases and cancer predisposition. Alterations in TAD boundaries and higher-order chromatin structure have already been implicated in developmental disorders²⁸. Systematic generation of haplotype-resolved T2T assemblies with chromatin topology across diverse populations may therefore provide a more complete reference framework for interpreting pathogenic variation.

At the same time, clinical translation will require further improvements in cost efficiency, turnaround time, and standardized analysis pipelines. Although our workflow reduces the need for multiple sequencing platforms, UL DNA preparation remains technically demanding and sensitive to sample quality. Continued protocol optimization and automation will be critical for broader adoption.

In summary, we present a scalable, Nanopore-only workflow for diploid T2T genome assembly that integrates sequence, methylation, and chromatin topology from four flow cells per individual. By lowering technical and infrastructural barriers, this approach facilitates expansion of T2T genomics beyond large consortia and supports population-scale and functional genomic studies. Ongoing efforts to expand this resource to 50 publicly available T2T genomes will further contribute to the development of comprehensive pangenome references and functional diploid genome maps.

Methods

Cohort description

Genomes of 23 healthy adults were assembled, representing diverse ancestries including European (n=18), Latin American (n=2), South Asian (n=1), and East/Southeast Asian (n=2) backgrounds. Continental groups were assigned based on the reported country of birth by participants (Supplementary Table 1). The cohort comprised 11 females and 12 males (mean age 45.9 years; range 25–95 years). Several participants reported mixed ancestral origins. Two individuals (T2T00 and T2T04) were sequenced together with their parents to enable trio-based phasing. Participants were

384 recruited by open invitation at Institute of Medical Genetics and Applied Genomics, University of
385 Tübingen. All participants were carefully counselled on the implication of open access reference data
386 by one of us (OR). Informed consent was obtained from all participants for use of their samples for
387 research and of the data for T2T, as approved by the ethics committee of the University Hospital
388 Tübingen in accordance with the Declaration of Helsinki (GeHealth ethics approval No.
389 113/2025BO1).

390

391 **DNA extraction and ONT UL Sequencing**

392 Ultra-high molecular weight (UHMW) DNA was extracted from fresh EDTA blood, collected in EDTA
393 anticoagulant tubes, using the NEB High Molecular Weight DNA Extraction Kit for Cells & Blood (New
394 England Biolabs, T3050) according to the manufacturer's protocol with optimizations for UL read
395 sequencing. The UHMW DNA was eluted in ONT Extraction Elution Buffer (EEB) and incubated at
396 room temperature for several days to achieve complete homogenization before library preparation.
397 For each individual, two UL libraries were prepared from 30–40 µg of unsheared DNA eluted in 750
398 µL of EEB buffer using the ONT UL Kit V14 (SQK-ULK114). Libraries were pooled and sequenced across
399 three PromethION R10.4.1 flow cells (FLO-PRO114M) at a total loading volume of 150 µL per load (75
400 µL SBU, 7.5 µL LSU, 67.5 µL library). Flow cells were periodically washed and reloaded with fresh
401 libraries up to 5 times when active pore counts declined below ~25% to maximize throughput.

402 The HG002 cell line (GM24385; Coriell Institute for Medical Research) was cultured according to the
403 manufacturer's protocol in RPMI medium (Sigma-Aldrich) supplemented with 2 mM L-Glutamine,
404 100 U/mL penicillin, 100 µg/mL streptomycin, and 10% heat-inactivated fetal calf serum (FCS;
405 Thermo Fisher Scientific) at 37 °C in a humidified atmosphere containing 5% CO₂. UHMW DNA was
406 extracted from the cultured cells using the same extraction workflow described above. Two UL
407 sequencing libraries were prepared from the extracted DNA, and UL sequencing was subsequently
408 performed as described above.

409

410 **Pore-C Library Preparation and Sequencing**

411 Peripheral blood mononuclear cells (PBMCs) were isolated from fresh peripheral blood using Ficoll
412 density-gradient centrifugation and processed following the Pore-C protocol. Briefly, 20 million cells
413 were crosslinked with 1% formaldehyde for 10 minutes, digested overnight with restriction enzyme

414 NlaIII, and proximity ligated using T4 DNA ligase for 6 hours. Subsequently, proteins were digested
415 overnight with proteinase. Ligated genomic DNA was isolated and purified using self-prepared SPRI
416 beads. After reverse crosslinking and purification, 2.4 µg DNA was prepared using the ONT Ligation
417 Sequencing Kit V14 (SQK-LSK114 XL) with modifications to the manufacturer's protocol aimed at
418 enhancing efficiency. The DNA was further processed in a series of steps, including nick repair and
419 end preparation, to facilitate the attachment of sequencing adapters. Pore-C libraries were
420 sequenced on PromethION R10.4.1 flow cells (typically 1-2 per sample; five for selected samples).
421 Flow cells underwent multiple nuclease washes and reload cycles.

422

423 **Basecalling, Error Correction, and Read Filtering**

424 All raw data were basecalled using *Dorado v1.0.2* with the R10.4.1 super-accuracy (SUP) model
425 (dna_r10.4.1_e8.2_400bps_sup@v5.2.0). 5mC and 5hmC modifications were called using
426 5mCG_5hmCG@v1. **UL** reads were error corrected using *dorado correct* (*herro-v1* model). These
427 corrected reads were used as high-quality (HQ) input reads in place of PacBio HiFi reads for assembly.
428 Read statistics were computed using *FastCAT v0.15.2* and *dorado summary*. Reads were filtered and
429 subsampled using *filtlong* to 50x coverage for HQ reads and 70x for UL reads. Minimum read length
430 thresholds were 10 kb (HQ) and 80 kb (UL). A minimum read quality score of 9 was applied. We used
431 the parameter *--length_weight 10* for *filtlong* to favor length over quality in read subsampling.
432 For genome assembly, we used a single Pore-C flow cell for which we copied the FASTQ files to the
433 assembly input folder. For *Verkko* assembly this file was used without preprocessing, while a pseudo-
434 paired-end Hi-C format was generated for the *hifiasm* assembler using a custom python script. To
435 this end, Pore-C reads were split at CATG motifs and the resulting fragments combined with up to 5
436 other fragments of the same read into a set of pseudo-paired-end reads.

437

438 **Genome assembly**

439 **Verkko Assembly**

440 Genome assemblies were generated using *Verkko v2.2.1*²⁹ in local mode on a single compute node
441 per sample. Corrected HQ reads were supplied as *--hifi* input and **UL** reads as *--ont*, with correction
442 disabled (*--no-correction*). To compare *Verkko* results with different phasing options, assemblies

443 were generated in: (1) Unphased mode; (2) Pore-C phasing mode (*--porec*); (3) Trio-aware mode using
444 haplotype-specific k-mers generated with *Meryl* and *Merqury*.

445 For trio binning with *Verkko*, we first counted homopolymer-compressed k-mers (k=30) for each
446 parent and child using *meryl count compress*. Then, we extracted haplotype-specific k-mers for each
447 parent using *hapmers.sh* from the *Merqury* package. These parental k-mer sets were used as input
448 for *Verkko*'s *--hap-kmers* option enabling trio-aware scaffolding and phasing.

449

450 **Hifiasm Assembly**

451 For samples T2T00 and T2T04, trio-based and Pore-C-based assemblies were generated using *hifiasm*
452 *v0.25.0-r726*⁹ with ONT error correction enabled (*--ont*) and telomere motif detection (*--telo-m*
453 *CCCTTA*). *Hifiasm* assembly with Pore-C phasing was performed using the parameters: *--telo-m*
454 *CCCTTA --dual-scaf --ul-cut 70000 -l2 --ont --h1 {porec_r1} -h2 {porec_r2} {ul}*.

455 For *hifiasm* trio binning of the samples T2T00 and T2T04, we generated k-mer databases (k=37) for
456 each parent using *yak count*. The resulting .yak files were supplied to *hifiasm* with the -1 and -2
457 options for maternal and paternal k-mers. *Hifiasm* with trio binning was performed using the
458 parameters: *--telo-m CCCTTA --dual-scaf --ul-cut 70000 -l0 --trio-dual --ont -1 {maternal} -2 {paternal}*
459 *{ul}*.

460

461 **T2T Classification**

462 Classification of chromosome sequences as T2T-contig or T2T-scaffold is determined by the presence
463 of telomeric motifs, full alignment to a reference chromosome and the presence of gaps (stretches
464 of NNN sequence). We classified chromosomes as T2T-Scaffolds if: (1) Telomeric repeats were
465 present at both ends. The telomere motif is TTAGGG (6-mer), and at least 10 tandem copies are
466 required at each end. (2) A single colinear alignment chain to T2T-CHM13 spanned ≥95% of the
467 assembled sequence and ≥95% of the reference chromosome. If T2T-Scaffolds were also gap free (no
468 N-stretches) they were classified as T2T-Contigs. A custom python script was used to identify and
469 count N-stretches in the assembled sequences and determine the final classification. The script
470 provides summary tables showing the number, size, and location of gaps.

471 To compute T2T-CHM13 coverage, we created a pairwise alignment in PAF format using *minimap2*
472 *v2.28* and then used *minigraph cov_cal* to identify sequences meeting both ≥95% criteria. *Cov_cal*

473 builds a colinear alignment chain per contig–chromosome pair and evaluates the span of that chain,
474 not the number of aligned bases. Insertions or deletions relative to the reference within an otherwise
475 continuous chain therefore do not cause the criterion to fail, making the classification tolerant to
476 inter-individual length variation.

477

478 **Assembly quality control**

479 To benchmark assembly continuity and completeness against current reference quality T2T
480 assemblies, we compared our genomes to 20 diploid assemblies from HPRC Release 2
481 (Supplementary Table 5). The assembly index (`assemblies_pre_release_v0.6.1.index.csv`) was
482 obtained from the HPRC intermediate assembly repository ([https://github.com/human-](https://github.com/human-pangenomics/hprc_intermediate_assembly)
483 [pangenomics/hprc_intermediate_assembly](https://github.com/human-pangenomics/hprc_intermediate_assembly), commit 41aa47d, 14 February 2025), and assemblies
484 were downloaded according to the S3 paths listed therein.

485 *Assembly Integrity*: Potential mis-assemblies were identified using *NucFlag* v0.3.3 and *Flagger* v1.1.0,
486 based on HQ read re-alignments. Error corrected HQ reads were re-aligned to the assembled
487 genomes using *minimap2* with the parameter preset *lr:hqae*. Mapped reads were analyzed with
488 *NucFlag* using default options. Plots were created for all regions using the Repeatmasker annotation.
489 *Flagger* was executed from a docker container in a two-step process. First, *flagger bam2cov* was used
490 to create a coverage file from the re-alignment. Second, *hmm_flagger* was used to generate a bed
491 file with potential integrity issues. Region flagged by *Flagger* and *NucFlag* were classified in assembly
492 coordinates using the per-sample *RepeatMasker* annotation into three mutually exclusive classes by
493 *bedtools*: satellite (*RepeatMasker* Satellite entries), other repeat (any other annotated repeat, not
494 satellite), and non-repetitive (no annotation).

495 *Genome completeness*: We assessed the completeness and quality of our T2T assemblies by
496 comparing against the T2T-CHM13 reference genome¹. Ensembl release 113 cDNA sequences were
497 aligned using *minimap2* with options *-cxsplice* and *-C5* to both the assembled genome and to T2T-
498 CHM13. The number of partial, complete and duplicate occurrences of every transcript were
499 assessed using *paftools asmgene* as part of *minimap2* v2.28. Genome completeness was then
500 calculated by dividing the sum of complete and duplicate alignments in the asmgene output by the
501 sum of full alignments in the reference. Furthermore, the numbers of missing multi copy genes
502 (MMC) and missing single copy genes (MSC) were calculated using *paftools asmgene*. MMC is

503 calculated by finding the number of multi copy genes in the assembly divided by the number of multi
504 copy genes found in the reference. MSC is the fraction of single copy genes found in the assembly to
505 the number of single copy genes found in the reference.

506 *Consensus quality:* To determine QV values we first performed short read WGS (srWGS) for all samples
507 using Illumina NovaSeq X+. QV values were then generated using *Merqury v1.3* by comparing k-mers
508 from the srWGS data to k-mers in the assembled genomes. QV values were calculated for three kmer
509 lengths (k=21, k=31, k=37) for each generated assembly. *Merqury* QV values are reported for all 23
510 cohort samples and are used for relative comparison between assemblies. Reference-based QV values
511 (see below), computed against the GiaB HG002 reference assembly, cover repeat contexts largely
512 invisible to k-mer comparison, and are therefore not directly comparable in magnitude.

513

514 **Mendelian Inheritance evaluation**

515 Mendelian consistency was evaluated for the two trios T00 and T004. Per-sample phased VCFs
516 (assembly to CHM13v2.0) were merged with *bcftools* merge. Evaluation was restricted to biallelic
517 heterozygous child sites with exactly one contig per allele (FORMAT/GT 0|1 or 1|0, AD 1,1), so that
518 each position is spanned by exactly two contigs. Consistency was assessed with *bcftools* and
519 *mendelian2*. Violations were classified as switch or flip (ALT in at least one parent) or *de novo* (ALT in
520 neither parent). Mendelian QV was computed as $-10 \log_{10}(n_{\text{de novo}} / 6e9 \text{ bp})$.

521

522 **Parent-of-origin assignment**

523 Parent-of-origin was assigned with an adapted implementation of *PatMat* (v1.4.0)²⁴. Assembly
524 derived phased small variants were restricted to diploid chromosomes with *bcftools*. Because the
525 Verkko and Pore-C assemblies are chromosome-scale phased, the phased VCF was supplied to
526 *PatMat* both as the per-variant input (-v) and as the chromosome-scale phase reference (-stv),
527 replacing the strand-seq input of the original method. *PatMat* was run against haplotagged UL
528 alignments and a set of imprinted DMRs. Assignment was determined for autosomes only.
529 Chromosome X was not assigned, as no imprinted DMRs are annotated and X-inactivation dominates
530 the methylation signal. Contigs were renamed in PanSN format encoding parent-of-origin.

531

532 **Independent evaluation against the GIAB Q100 HG002 benchmark**

To perform an independent quality benchmark of our approach, we generated new sequencing data for HG002 (3 PromethION R10.4.1 UL flow cells and 1 Pore-C flow cell) and assembled the HG002 genome using our T2T-ONT workflow described in this paper. The resulting assembly was compared to the GIAB Q100 HG002v1.1 benchmark assembly using *GQC v0.1* (<https://github.com/nhansen/GQC>). *GQC* aligns the query assembly to the benchmark, reports continuity statistics relative to the benchmark, identifies base-level discrepancies within benchmark-confident regions, and distinguishes consensus errors from phasing errors (correct variant assigned to wrong haplotype). Accuracy of mononucleotide runs and indel error rate as a function of indel length were taken directly from the *GQC* output to generate Supplementary Fig. 14. Errors reported by *GQC* were annotated using the GIAB v4.2 genome stratifications for HG002³⁰. Because *GQC* reports error coordinates in the HG002v1.1 haplotype-resolved coordinate system (chr*_MATERNAL / chr*_PATERNAL), the haplotype-specific stratification BED files were used directly rather than lifting coordinates to a linear reference. For each stratum, maternal and paternal BED files were merged with *bedtools merge*, and error positions were intersected with the merged intervals using *bedtools intersect*. Coding sequence annotations were taken from the JHU Liftoff v0.6 RefSeq liftover onto HG002v1.1. rDNA loci were extracted from the cenSat v2.0 annotation and are reported for completeness but are subsumed under the satellite category in all summary statistics, as they represent a structurally distinct subset of satellite arrays rather than an independent error class. Each error was assigned to exactly one category according to the priority order: coding > highly polymorphic (MHC, KIR, VDJ) > satellite > segmental duplication > tandem repeat > homopolymer > non-repetitive. This hierarchy reflects both functional importance and the reliability of the benchmark in each context, except for the non-repetitive category. Assembly quality values ($QV = -10 \log_{10}(e/b)$, where *e* is the number of errors and *b* the number of covered benchmark bases), were computed for the region categories: 1) all (combined), 2) non-repetitive, and 3) coding. The total length of the non-repetitive category was derived as total benchmark length minus the union of the length of coding sequence and all repeat strata.

Variant calling and phasing accuracy

UL reads were aligned to T2T-CHM13 v2¹ using *minimap2*³¹. Sex was inferred from X/Y coverage ratios (>0.5 results in male). Coverage profiles were calculated using *mosdepth*³² v0.3.8. Fully phased

single nucleotide variants and indels were called with *dipcall*³³ v0.3, larger structural variants with *hapdiff*³⁴ v0.9. Phasing accuracy was evaluated using *whatshap compare* against the GIAB CHM13 phased benchmark (CHM13v2.0_HG2-T2TQ100-V1.1²). We used *whatshap stats*³⁵ (v2.3) to generate variant statistics. Further variant statistics were extracted from VCF files generated by *hapdiff* and *dipcall* using custom python scripts.

Ancestry analysis

Variants were normalized using *bcftools norm* (v1.21) and intersected with a lifted 1000 Genomes panel in T2T-CHM13 coordinates using *bcftools view -T*, as suggested by Lalli et al.³⁶. Normalized VCFs were filtered to retain only biallelic SNVs and indels, excluding spanning deletions and large variants exceeding 50 bp in either reference or alternate allele length. Sample and reference VCFs were merged using *bcftools merge* with options *--merge none* to prevent forced merging of incompatible variants, and *--missing-to-ref* to impute missing genotypes as homozygous reference calls. The merged VCF was converted to PLINK2 v2.0.0a.6.9 binary format using *plink2 --vcf*. Chromosome codes were standardized with *--split-par b38* to handle pseudoautosomal regions on chromosome X, and half-calls were set to missing (*--vcf-half-call m*). Variant IDs were standardized using the format chr:pos (*--set-missing-var-ids @: #*). Quality control filters were applied to the merged dataset using *plink2* with the following lenient thresholds optimized for ancestry analysis: minor allele frequency (MAF) ≥ 0.01 to retain rare variants informative for population structure, per-variant missingness $\leq 15\%$ (*--geno 0.15*), and Hardy-Weinberg equilibrium p-value ≥ 0.001 (*--hwe 1e-3*), with the latter threshold intentionally lenient to accommodate population structure and potential familial relationships. Variant IDs were updated to a standardized format (chr:pos:ref:alt) to ensure consistency across the dataset.

PCA was performed using PLINK2 following LD pruning ($r^2=0.2$). Ancestry inference was performed using *RFMix*³⁷ v2.03. Variants in strong linkage disequilibrium were removed using *plink2 --indep-pairwise* with a 50-SNP sliding window, 10-SNP step size, and r^2 threshold of 0.2. PCA was performed using *plink2 --pca* to extract the top 20 principal components. The resulting eigenvectors and eigenvalues were used to visualize population structure, stratified by either population or super-population labels of 1000GP. We used *RFMix* to identify local and global ancestry of the subjects with parameters *crf-spacing=0.001* and *rf-window-size=0.1*.

593

594 **Haplotype-phased 3D genomes using Pore-C**

595 Pore-C data were processed using *wf-pore-c*³⁸ **v1.3.0** and *pairtools*³⁹ **v1.1.3** to generate contact maps
596 (pairs files). Haplotype tagging of Pore-C reads was performed as described by Chen et al.¹⁹. In brief,
597 reads were mapped to the reference genome using *Falign* **v2.0.0**. SNVs and indels in high-quality
598 regions with $\geq 5\times$ Pore-C coverage were used for *haplotagging* with *dip3d haplotag* (**v2.0.0**). Inter-
599 chromosomal contacts were not included in the phasing. Finally, separate alignment files for each
600 haplotype were extracted. Haplotype-separated Pore-C alignment files were processed into pairs
601 format using *pairtools parse2*³⁹. To count correct non-adjacent contacts on the same fragment we
602 used the *--expand* and *--flip* options. Contact maps from multiple flow cells of the same sample were
603 merged with *pairtools merge*. Statistics were calculated with *dip3d stats*. Juicebox⁴⁰ **v2.20.00** was
604 used for visualization of contact maps.

605 For further analysis we converted the pairs files to *Cooler*⁴¹ format using *cooler cload pairs* (**v0.10.3**)
606 with a minimum binwidth of **1,000** and then created a mcool file using *cooler zoomify* with aggregate
607 resolutions for binwidths **1,000, 5,000, 10,000, 50,000, 100,000** bp. Finally, we created a corrected
608 matrix using the ICE algorithm with cooler balance that serves as basis for detection of 3D features.
609 We used *cooltools insulation* (v0.7.1) to calculate insulation scores for the resolutions 25000bp and
610 40000bp and *cooltools eigs-cis* to create a compartment track. For TAD and Loop calling we used
611 *hicExplorer* tools⁴² v3.7.6 again starting from the balanced pairs file. Then we used *hicFindTADs* to
612 create a TAD score and Boundary plot for the resolutions.

613

614 **Haplotype-resolved Methylation Analysis**

615 Modification-aware UL reads were subsampled to a maximum size of 100 Gigabytes and aligned to
616 both the assembled genomes and the T2T-CHM13 reference genome with minimap2. Reads were
617 haplotagged with phased SNP calls using *longphase haplotag* (**v2.0**). Reads that could not be
618 confidently assigned remained untagged (HP:0). Haplotagged reads were split into separate BAM
619 files for each haplotype using *samtools view -d HP*. Haplotype-specific methylation frequencies were
620 computed with *modkit* **v0.5.1** and exported as bigWig files. In brief, case modification consensus was
621 generated using *modkit pileup*, summary statistics were generated by *modkit summary*, and
622 bedMethyl files were converted to bigWig format using *modkit bedmethyl tobigwig*, enabling

623 visualization of CpG methylation in genome browsers on both T2T-CHM13 and T2T-sample
624 coordinates.

625

626 **Centromere and telomere structures**

627 Centromeric regions were characterized using *CenMap* v1.1.1 tools with default settings on all 23
628 samples as described in Logsdon et al.⁴. *CenMap* finds satellite repeats, annotates the High Order
629 Repeat (HOR) classification, identifies centromeric dip regions (CDRs), and assesses assembly
630 correctness of the centromeres using *NucFlag*. We visualized the HOR structure using *cenplot* in a
631 custom configuration.

632

633 **Data Availability**

634 The data generated in this study along with the assembled genomes are available via controlled
635 access in the German Human Genome-Phenome Archive (GHGA, data.ghga.de) under the GHGA
636 Accession <https://data.ghga.de/study/GHGAS08644233257767>. Further details, including the data
637 access policy for the study, can be found there. Genome assemblies, additional annotations and
638 further analyses results are available via open access at https://github.com/imgag/T2T_Genomes. All
639 code used for presented analyses is available at https://github.com/imgag/T2T_ONT.

640

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652

653 **Author Contributions**

654 O.R., S.O., B.K., and C.G. designed and led the study. R.P., F.C., H.S.H., E.B.A. performed sample
655 extraction, library prep and sequencing. C.G. and S.L. implemented the computational analysis
656 pipeline and performed data analyses. N.C. carried out resource management. C.G., S.O., O.R. wrote
657 the main manuscript, with revisions by all authors. All authors have reviewed and approved the final
658 manuscript.

659

660 **Use of AI and language tools**

661 All authors confirm that this manuscript was conceived, prepared, and written by the authors. The
662 content reflects the original work, ideas, and analyses of the authors. DeepL and Claude were used
663 for grammar and language corrections. Claude was used to assist in code writing and documentation.
664 All text and code were reviewed and validated by the authors, who take full responsibility for their
665 content and correctness.

666

667 **Competing Interests**

668 The authors declare no competing interests.

669

670 **List of Abbreviations**

- 671 • 3C Chromatin conformation capture
- 672 • α SAT alpha satellite
- 673 • bp base pairs
- 674 • CDR Centromeric Dip Regions
- 675 • DNA Deoxyribonucleic Acid
- 676 • GHGA German Human Genome Archive
- 677 • GIAB Genome in a Bottle
- 678 • HOR high-order repeat
- 679 • HQ High Quality
- 680 • HPRC Human Pan-Genome Reference Consortium
- 681 • kb kilobases
- 682 • MMC missing multi copy gene

- 683 • MSC missing single copy gene
- 684 • ONT Oxford Nanopore Technologies
- 685 • QC Quality Control
- 686 • SNV Single Nucleotide Variant
- 687 • srGS short read WGS
- 688 • SV Structural Variants
- 689 • T2T Telomere-to-Telomere
- 690 • TAD Topologically Associating Domain
- 691 • UHMW Ultra high molecular weight
- 692 • UL Ultra Long Nanopore Reads
- 693 • WGS whole genome sequencing
- 694 • QV Quality values

695

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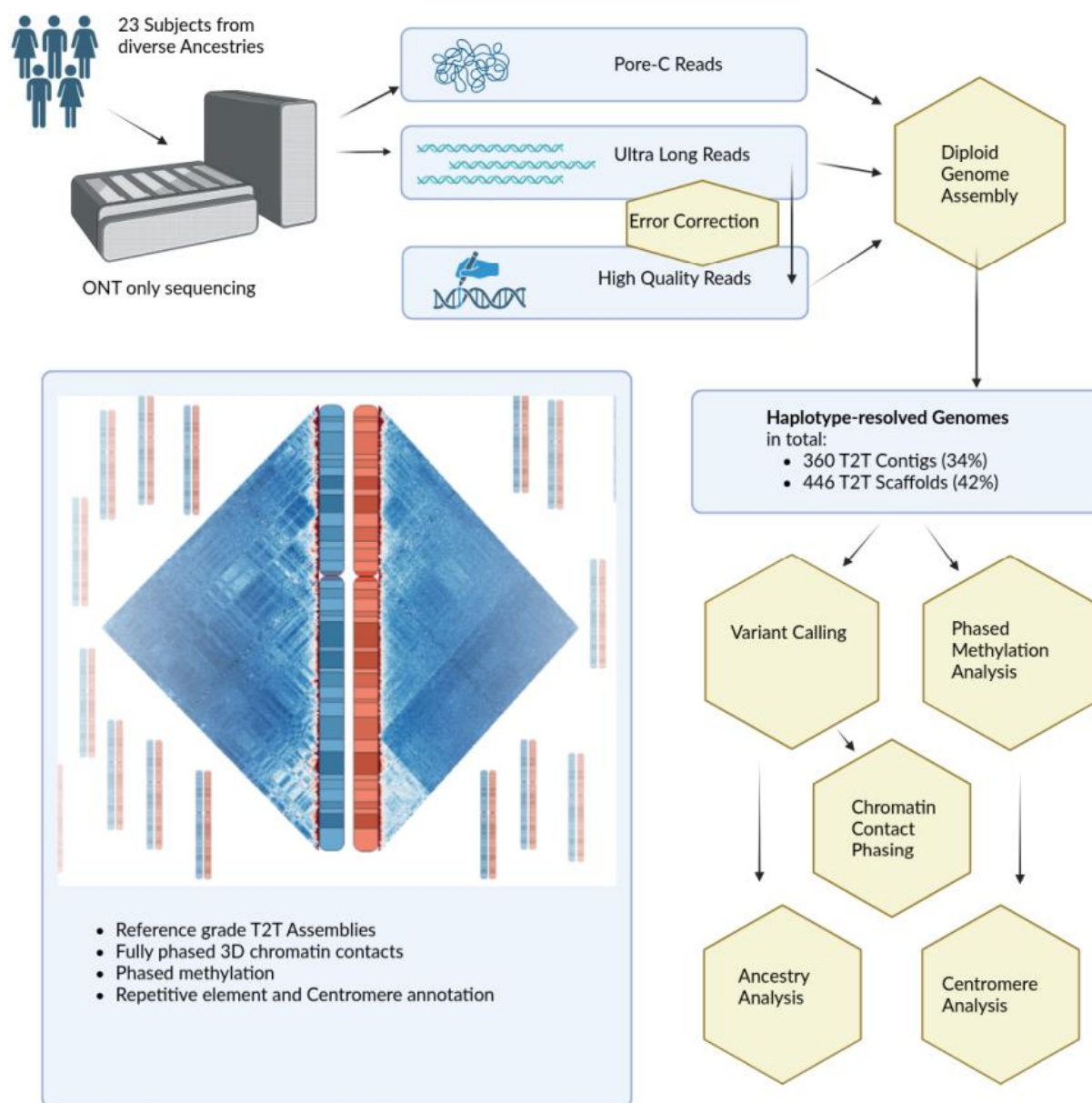
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826

827 **Figure 1: Graphical abstract.**

828 Graphical overview of the experimental and computational workflow. Fresh peripheral blood is
 829 processed to generate ultra-long (UL) Nanopore libraries and Pore-C libraries. UL reads are error-
 830 corrected to produce high-quality (HQ) reads for graph-based de novo assembly using Verkko. Pore-
 831 C data enables chromosome-scale scaffolding and haplotype phasing. Downstream analyses include
 832 assembly quality control, phased variant detection, centromere characterization, haplotype-resolved
 833 methylation profiling, and 3D chromatin contact map generation. Our standardized T2T-3D procedure
 834 integrates all steps into a unified workflow.

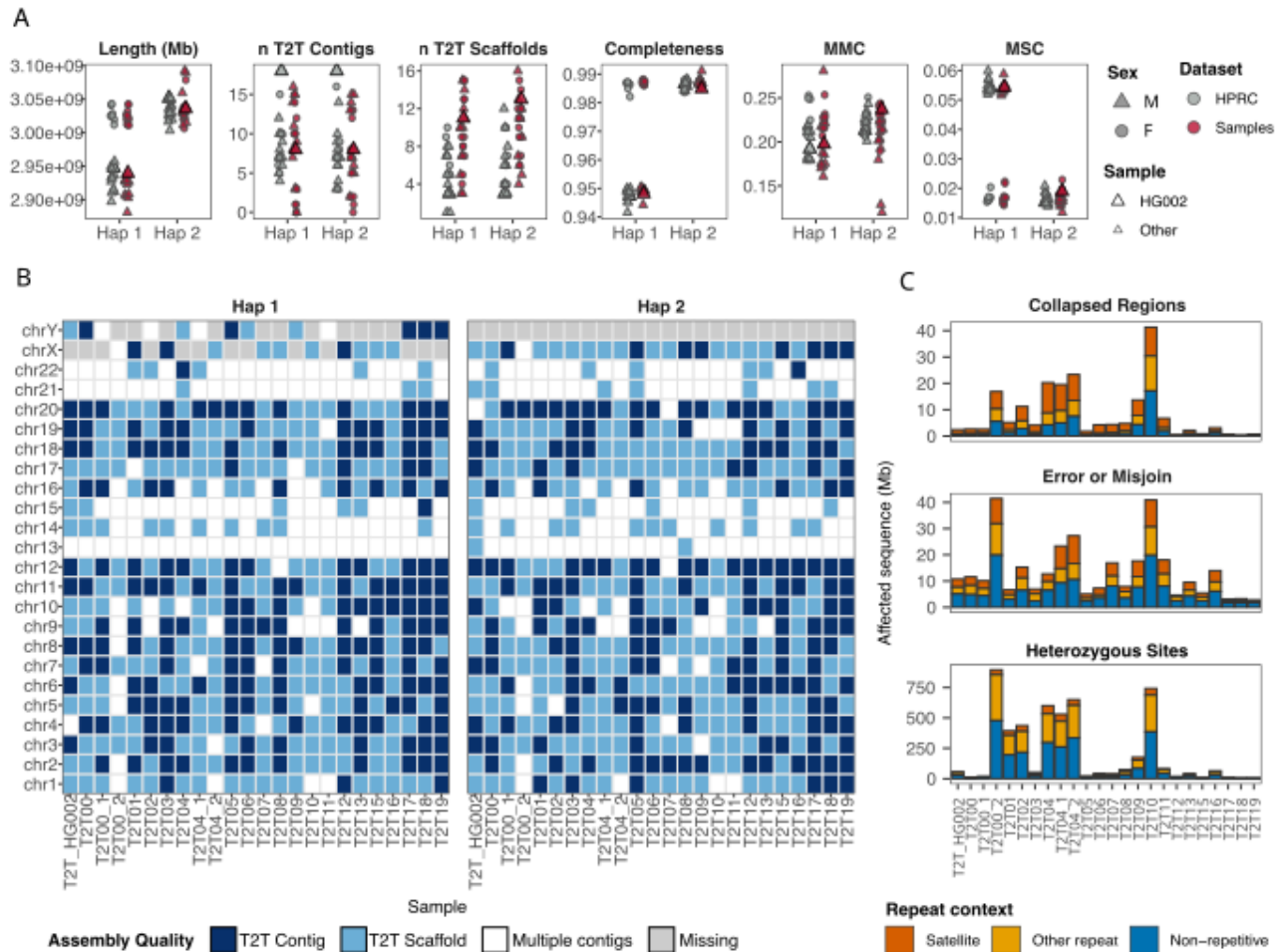
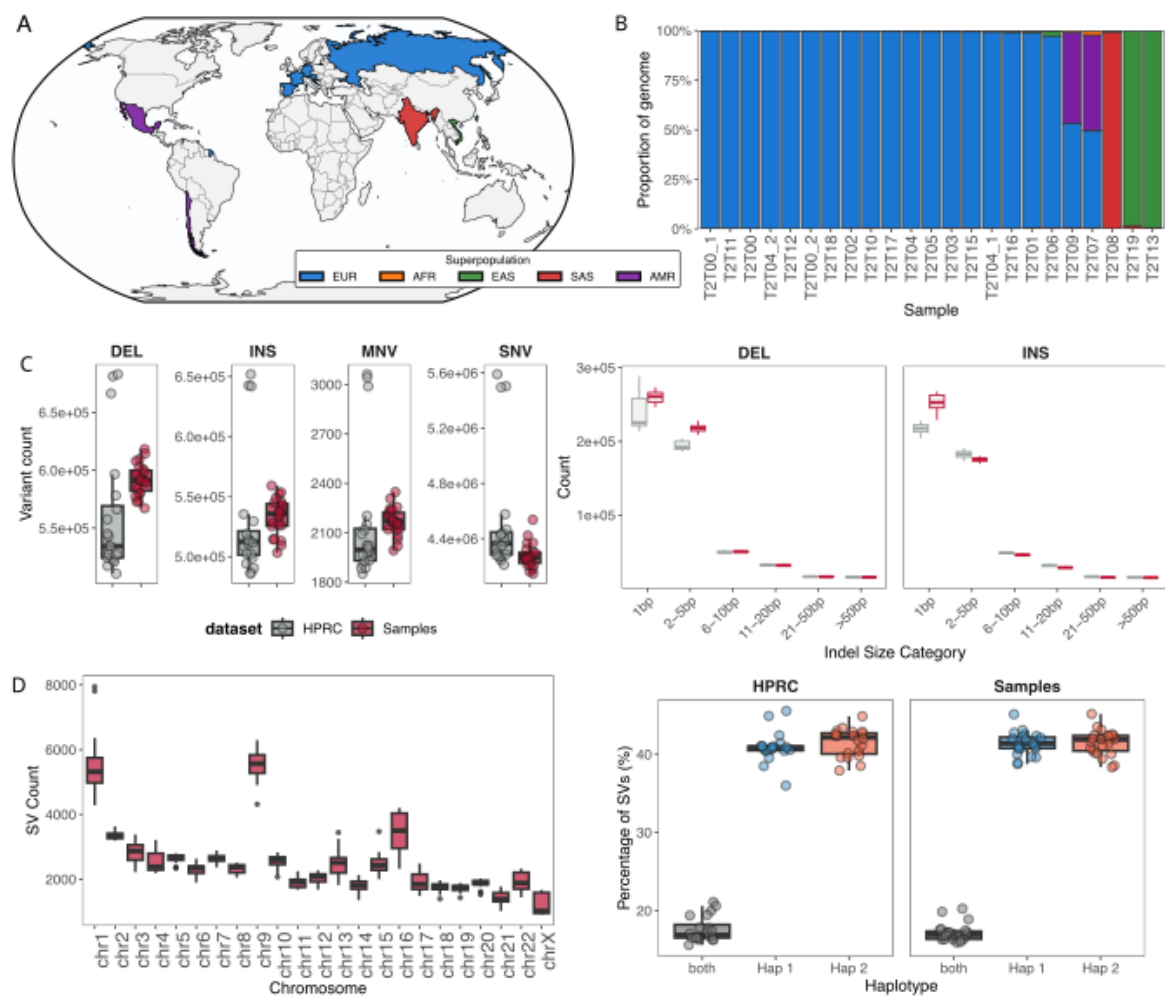


Figure 2: Nanopore-only assemblies achieve hybrid-grade T2T quality metrics.

(A) Comparison of 23 Nanopore-only assemblies (red) with 20 multi-platform HPRC assemblies (gray) using identical quality control metrics. Shown are assembled genome length, number of gapless T2T contigs, number of near-complete T2T scaffolds, gene completeness, fraction of missing multi-copy genes (MMC), and fraction of missing single-copy genes (MSC). Distributions are comparable across datasets. **(B)** Chromosome-level assembly continuity across both haplotypes for all individuals. Dark blue indicates gapless T2T contigs; light blue indicates near-complete T2T scaffolds containing ≥ 1 gap; white indicates incomplete chromosomes; grey indicates missing sex chromosomes. **(C)** Potential assembly issues detected by NucFlag following realignment of HQ reads to assembled genomes. Colors indicate the underlying repeat context and each bar represents an individual sample, including HG002 and the T2T assemblies; Variability reflects differences in sequencing and library preparation performance across samples.



850 **Figure 3: Phased variant discovery and ancestry validation in diploid T2T genomes.**

851 **(A)** Geographic origin of the 23 study participants, grouped by 1000 Genomes *continental group*
852 (European [EUR], East Asian [EAS], South Asian [SAS], Admixed American [AMR]). **(B)** Ancestry
853 inference of the assembled genomes, performed with RFMix using the 1000 Genomes Project
854 reference panel. **(C)** Small variant counts in Nanopore-only assemblies (red) and 20 HPRC assemblies
855 (gray), including single nucleotide variants (SNVs), insertions (INS), deletions (DEL), and multi-
856 nucleotide variants (MNVs). Right panel shows size distributions of insertions and deletions. **(D)**
857 Structural variant (SV) distribution across chromosomes, summed over all samples (left). Right panel
858 shows allelic distribution of heterozygous and homozygous SVs in the study cohort compared to HPRC
859 samples.

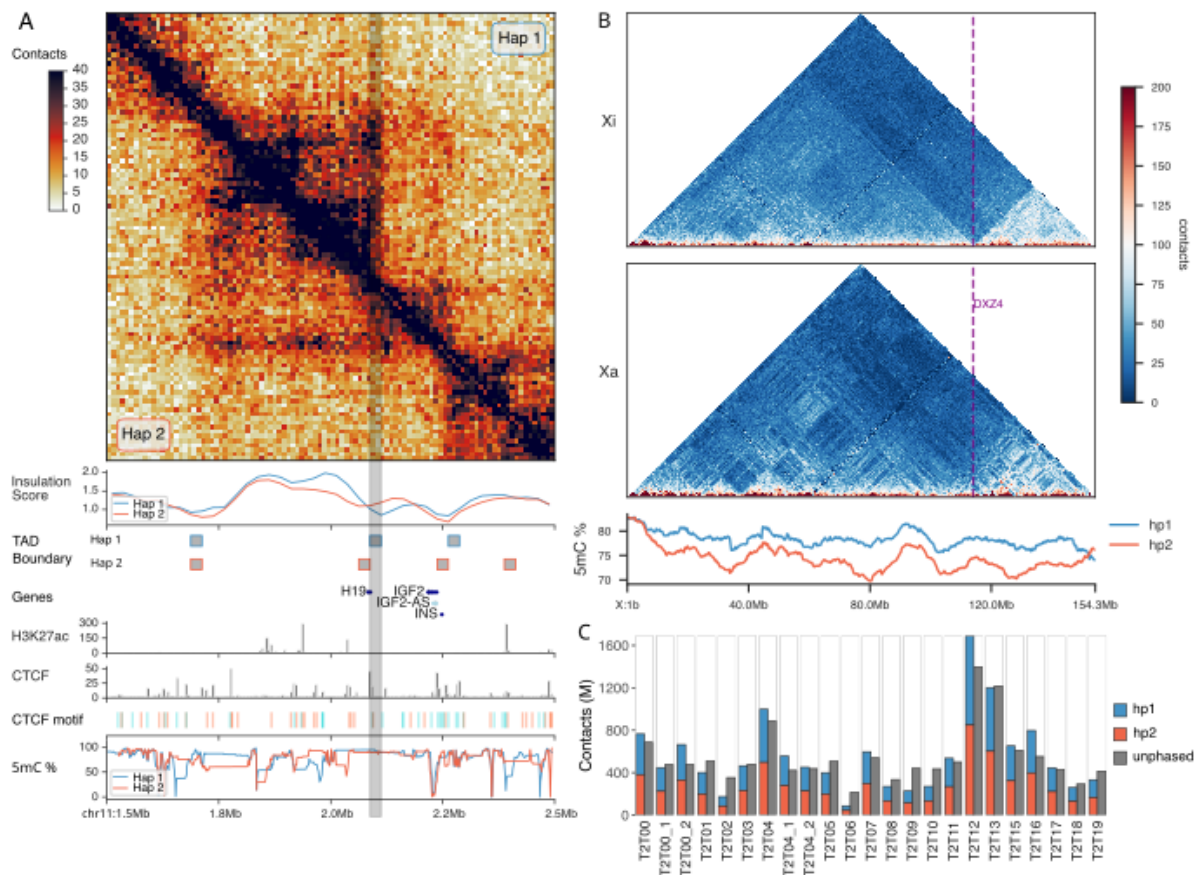
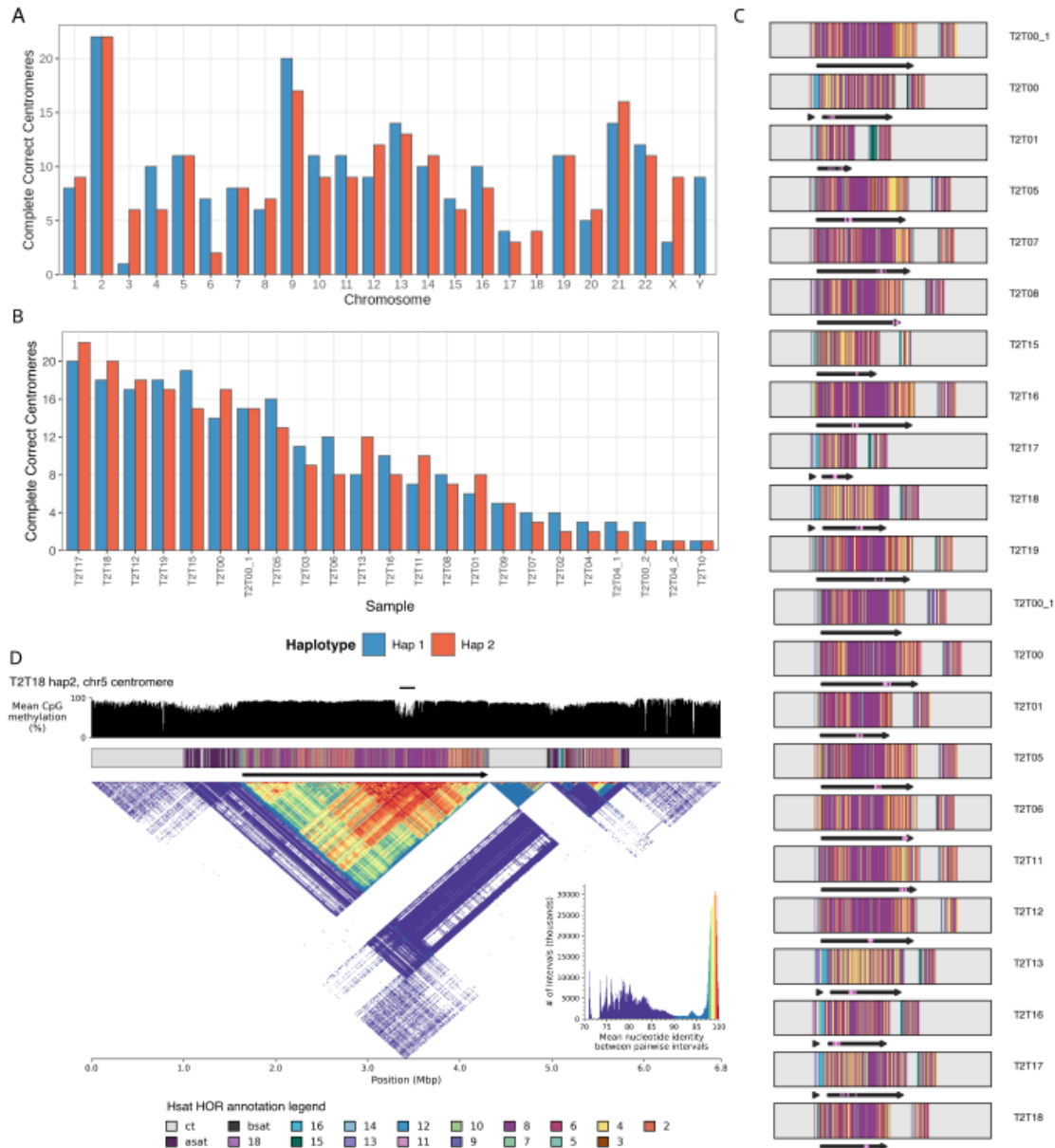


Figure 4: Haplotype-resolved Pore-C reveals allele-specific 3D chromatin organization.

(A) Haplotype-specific chromatin contact maps (10 kb resolution) for the imprinted IGF2-H19 locus (chr11:1.5–2.5 Mb) in sample T2T12. Insulation scores highlight topologically associating domain (TAD) boundaries. Tracks include RefSeq gene annotation, H3K27ac signal, CTCF binding sites and motifs, and smoothed CpG methylation frequency (5mC). Distinct haplotype-specific chromatin structures are evident. **(B)** Chromosome-wide haplotype-separated contact maps (100 kb resolution) for chromosome X in T2T12, demonstrating differences between active and inactive X chromosomes. Bottom track shows smoothed CpG methylation frequency. **(C)** Number of chromatin contacts per sample. Unphased contacts were generated using wf-pore-c; haplotype-resolved contacts were obtained using a modified dip3d pipeline (cis interactions only).



873

874 **Figure 5: Nanopore-only assemblies reconstruct complete centromeric structures.**

875 **(A)** Number of fully resolved and structurally validated centromeres per chromosome across all 23

876 individuals and both haplotypes. **(B)** Total number of complete, validated centromeres per individual.

877 **(C)** Example centromere structure for chromosome 5 across individuals. Colors denote higher-order

878 repeat (HOR) classes. Black arrows indicate HOR arrays; centromeric dip regions (CDRs) are

879 highlighted in pink. **(D)** Repeat homology plot for chromosome 5 centromere (haplotype 2, sample

880 T2T18). Top track shows CpG methylation with CDR annotation. Second track shows HOR class

881 annotation. Heatmap indicates pairwise nucleotide identity within satellite repeat arrays.